MILITARY SPECIFICATION
INITIATOR, PROPELLANT ACTUATED, DELAY, JAU-3/A25 ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Propellant Actuated, Delay, JAU-3/A25, Assembly loaded with II-A pellets, an M42 primer and a delay element for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-H-25579	- Hose Assembly Tetrafluoroethylene, High Temperature Power Plant, Aircraft

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

DRAWINGS

Munitions Command

F11731674	- Initiator, Propellant Actuated, Delay, JAU-3/A25 Assembly
D9280891	- Box, Fiberboard (Inner), Packing, Ammunition for Initiator, Delay (Gas-Subminiature)
D9280892	- Box, Packing, Ammunition for Initiator, Delay (Gas-Subminiature)

FSC 1377

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D9280893	- Box, Fiberboard (Outer), Alternative, Packing Ammunition for Initiator, Delay (Gas-Subminiature)
IEL-11731674	- Inspection Equipment List for Initiator, Propellant Actuated, Delay, JAU-3/A25 Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General.-Materiels, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Breakaway torque.-Each threaded joint of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.3 Residual magnetism.-The initiator shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional.

3.4.1 Ballistics.-Initiators loaded with accepted, pyrotechnic components (II-A pellets, M42 primer and delay element) and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be recorded in an 0.062 cubic inch volume at the end of 15 feet of aircraft hose, size No. 4 as specified in MIL-H-25579.

TABLE I

Temperature	Pressure (PSI) Min.	Pressure (PSI) Max.	Ignition Delay, Max. (see 6.3)
-65° to -70°F	890	2300	1.020 sec., max.
65° to 75°F	915	2300	0.850 ± 0.085 sec.
200° to 205°F	1000	2300	0.680 sec., min.

3.4.1.1 Initiation.-The initiatoon shall be fired by a source delivering an output equal to that of an M3A2 Initiator at the end of 15 feet of aircraft hose, size No. 4, in accordance with MIL-H-25579 at 70 ± 5 F.

3.4.2 The initiator shall not fail to fire.

3.4.3 The initiator shall function without mechanical failure (see 6.2).

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3.5 Mechanical Properties.-Mechanical Properties required by the drawing shall be uniform throughout the part within specified tolerances.

3.6 Shear Pin Strength.-The shear pin after application of protective finish shall shear when a load of 41 ± 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.7 Protective finish.-Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour.

3.8 Hermetic seal.-The assembled initiator, less the firing mechanism shall withstand 14 PSI, internal, external pressure differential with a maximum permissible leak rate of 1×10^{-4} cubic centimeters per second.

3.9 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Metal defects.-The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.9.2 Burr.-The part shall not have a burr which might be injurious to personnel handling the item.

3.9.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

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4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 21 assembled initiators for ballistic tests, 50 shear pins, and 10 sets of unassembled parts for visual and dimensional examination and tests.

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging shall be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 First article samples and ballistic acceptance tests samples of ballistic subassemblies shall be submitted in accordance with their applicable specification.

4.2.1.2 Examination and tests.-After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.3 Initial production failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with acceptable explosive and pyrotecnic components.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and associated criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group the criteria in Table III shall be used. All non-conforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The surface roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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4.3.2 (Cont'd)

TABLE II

Lot Size	Sample size	Major Accept	Major Reject	Minor Accept	Minor Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE III

Total Allowable Defects per Sample Size				
Sample Size for Major Defects	110	150	225	300
No. of Defect in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14
Sample Size for MInor Defects	**110**	**150**	**225**	**300**
No. of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 Pin, Firing, Dwg. C11738410 covering a detail of Dwg. C11731492

Defect	Method of Inspection
CRITICAL	
1. Diameter of striker	Gage
2. Length of striker	Gage
3. Metal defective	Visual
MAJOR	
101. True position of "0" ring groove with body	Gage
102. Concentricity of striker with body	Gage
103. True positon of shear pin hole with body	Gage
104. Diameter of "0" ring groove, min.	Gage
105. Diameter of body, min.	Gage
106. Diameter of shear pin hole, max.	Gage

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR (Cont'd)	
107. Depth of shear pin hole, max.	Gage
108. Width of "O" ring groove	Gage
109. Location of shear pin hole	Gage
110. Permissible variation of "O" ring groove	Gage
111. Spherical radius of tip	Gage
112. Protective finish inadequate or defective	Visual
113. Burr or sharp corner, except where permitted	Visual
114. Foreign matter	Visual
MINOR	
201. Diameter of "O" ring groove, max.	Gage
202. Diameter of body, max.	Gage
203. Diameter of shear pin hole, min.	Gage
204. Depth of shear pin hole, min.	Gage
205. Width of slot	Gage
206. Depth of slot	Gage
207. Total length	Gage
208. Length of body to "O" ring groove	Gage
209. Location of marking	Gage
210. Marking missing, incorrect or illegible	Visual
211. Surface finish improper, except as otherwise classified	Visual

4.3.2.2 Housing, Dwg. D11731622, covering a detail of Dwg. C11731492

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Pitch diameter of external thread, min.	Gage
102. Major diameter of external thread, min.	Gage
103. Pitch diameter of internal thread, max.	Gage
104. Minor diameter of internal thread, max.	Gage
105. True position of pitch diameter of external thread with end flange diameter	Gage
106. Concentricity of pitch diameter of internal thread on large end with center cavity	Gage

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
MAJOR (Cont'd)		
107.	True position of pitch diameter of internal thread with center cavity	Gage
108.	True position of striker hole with center cavity	Gage
109.	True position of shear pin hole with center cavity	Gage
110.	True position of flats with internal thread	Gage
111.	Perpendicularity of striker face with center cavity	Gage
112.	Perpendicularity of striker face with internal thread on large end	Gage
114.	Diameter of shear pin counterbore	Gage
114.	Diameter of shear pin hole centerbore	Gage
115.	Diameter of striker hole	Gage
116.	Diameter of bore, large end	Gage
117.	Depth of bore, large end	Gage
118.	Diameter of bore, small end	Gage
119.	Depth of bore, small end	Gage
120.	Depth of shear pin counterbore hole	Gage
121.	Length of shear pin hole	Gage
122.	Length of internal thread, on large end	Gage
123.	Length of internal thread on small end,min.	Gage
124.	Length of internal thread on small end, max.	Gage
125.	Width of thread undercut	Gage
126.	Location of shear pin hole	Gage
127.	Diameter of center cavity	Gage
128.	Width of striker plate	Gage
129.	Diameter of counterbore, small end	Gage
130.	Chamfer at mouth of small end	Gage
131.	Thread lead angle	Gage
132.	Surface finish of cavity	Gage
133.	Protective finish inadequate or defective	Visual
134.	Burr or sharp corner	Visual
135.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of external thread, max.	Gage
202.	Pitch diameter of internal threads, min.	Gage
203.	Width across flats	Gage
204.	Diameter of flange	Gage
205.	Width of flange	Gage
206.	Depth of counterbore	Gage
207.	Location of external thread from face	Gage
208.	Location of shoulder from face	Gage

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
209. Width of shoulder	Gage
210. Diameter of shoulder	Gage
211. Total length	Gage
212. Diameter of large end	Gage
213. Angle at bottom of bore, small end	Gage
214. Surface finish improper, except as otherwise classified	Gage

4.3.2.3 Pin, Shear, Dwg. B11731626, covering a detail of Dwg. C11731492

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Diameter, min.	Gage
102. Total length, min.	Gage
103. Surface finish improper	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter, max.	Gage
202. Total length	Gage

4.3.2.4 Bracket, Mounting Dwg. C11731624 covering a detail of Dwg. F11731674.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of internal thread, max.	Gage
102. Major diameter of internal thread, min.	Gage
103. Diameter of mounting holes	Gage

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR (Cont'd)	
104. Length of mounting holes centers	Gage
105. Height from base to thread hole center	Gage
106. Length from shoulder to thread hole center	Gage
107. Shoulder radius incorrect	Gage
108. Width of shoulder	Gage
109. Protective finish inadequate or defective	Visual
110. Burr or sharp corner	Visual
111. Foreign matter	Visual
MINOR	
201. Pitch diameter of internal thread, min.	Gage
202. Minor diameter of internal thread, max.	Gage
203. Total base length	Gage
204. Width of mounting surface (base)	Gage
205. Height from base to top of shoulder	Gage
206. Thickness of base	Gage
207. Length from center line to base end	Gage
208. Mounting hole location	Gage
209. Length of base relief	Gage
210. Depth of base relief	Gage
211. Base relief location incorrect	Gage
212. Angle of base relief incorrect	Gage
213. Radius, angles, or chamfers missing or incorrect	Gage
214. Surface finish improper	Visual

4.3.2.5 Delay Element Assembly 0.85 Sec. Delay Drawing F11731584 covering a detail of Dwg. F11731674. (see 6.6)

CRITICAL	
1. Improper assembly	X-ray 1/ and Visual
2. Any part missing	X-ray 1/ and Visual
3. Mixed ammunition	Visual
4. Flash hole missing or obstructed	Visual
5. Marking missing, incorrect or illegible	Visual
6. Metal defective	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Output charge not visable through flash hole	Visual
102. Primer retainer not flush or below surface "S"	Visual
103. Junction of primer retainer and delay body not properly sealed or crimped	Visual
104. Protective finish inadequate or defective	Visual
MINOR	
201. Varnish missing	Visual

1/ The delay element shall be X-Rayed with the long axis of the delay element perpendicular to the rays of the X-Ray machine. The X-Ray negatives, properly identified, shall be forwarded to the procuring agency.

4.3.2.6 Initiator M104 Assembly Drawing D10551446 covering a detail of Drawing F11731482. (see 6.6)

CRITICAL	
1. Mixed ammunition	Visual
2. Metal defective	Visual
3. Vent hole obstructed	Visual
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Thread damaged	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.3.2.7 Firing Mechanism Assembly, Drawing F11731492, covering a detail of Drawing F11731674.

CRITICAL	
1. Firing pin protrusion	Gage
2. Shear pin not engaged in firing pin	Gage
3. Striker hole in initiator housing obstructed	Visual

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. O-ring not lubricated	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.3.2.8 Assembly, Drawing F11731674

CRITICAL	
1. Any part missing	X-ray 1/ and Visual
2. Improper assembly	X-ray 1/ and Visual
3. Total length, max.	Gage
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
201. Thread sealing compound missing	Visual
202. Shipping cap missing	Visual
203. Protective finish inadequate or defective	Visual

1/ The assembly shall be x-rayed with the long axis of the initiator perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.9 Box, Fiberboard (Inner) Packing, Drawing D9280891.

CRITICAL

None defined

MAJOR

None defined

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4.3.2.9 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Cut or damaged through all impregnated layers	Visual
203. Marking missing, incorrect or illegible	Visual
204. Tape incomplete or badly wrinkled	Visual
205. Length of tear tab, min.	Visual
206. Color of tape incorrect	Visual

4.3.2.10 Box, Packing, Drawing D9280892

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.2.11 Box, Fiberboard (Outer) Alternative, Drawing D9280893.

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Contents loose	Manual
202. Carton broken or damaged	Visual
203. Number of units contained incorrect	Visual
204. Marking missing, incorrect or illegible	Visual
205. Tape incomplete	Visual

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4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Shear pin strength.-Fifty pins from each lot of shear pins tested by the contractor shall be tested for shear strength. Failure of any pin to comply with the requirements of 3.6 shall be cause for rejection of the lot.

4.3.3.2 Protective finish.-Unless otherwise specified, sampling plans and procedures in determination of the acceptability of coated parts and articles submitted by a supplier shall be in accordance with provisions set forth in MIL-STD-105 and MIL-C-68539.

4.3.3.3 Residual magnetism.-Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirement of 3.3 shall be cause for rejection of the assembly.

4.3.3.4 Breakaway torque.-Each initiator shall be subjected to the breakaway torque application. Failure of any initiator assembly to comply with the requirements of 3.2 shall be cause for rejection of the assembly.

4.3.3.5 Hermetic seal.-Each initiator shall be subjected to a hermetic seal test. Failure of any initiator to comply with the requirements of 3.8 shall be cause for rejection of the initiator.

4.3.3.6 Functional.-The following tests shall be performed by the Government:

4.3.3.6.1 Ballistics.-Twenty-one (21) units shall be tested to determine compliance with the requirements of 3.4.1, 3.4.2, and 3.4.3. The units shall be temperature conditioned as provided in Table IV and 4.4.7.1.1. Failure of any unit to comply with the requirements of 3.4.1, 3.4.2 and 3.4.3 shall be cause for rejection of the lot.

TABLE IV

Temperature	Sample Size
-65° to -70° F	7
65° to 75° F	7
200° to 205° F	7

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-11731674.

4.4 Test methods and procedures.

4.4.1 Shear pin strength.-After application of protective finish, the pin shall be subjected to the shear strength test to determine compliance with 3.6. The test shall be performed in the fixture in conjunction with a tensile testing machine.

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4.4.2 Protective finish.-Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.4.3 Breakaway torque.-A torque wrench set at the torque requirement specified in 3.2 shall be used in conjunction with the fixture to determine the adequacy of the joint. The torques shall be applied (1) between the female body and male body and (2) between the male or female body and the body and (3) between the body and the housing.

4.4.4 Residual magnetism.-The initiator shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects, by placing the initiator 6 inches from and at the same height as the compass in a north-south horizontal position (housing end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length. rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

4.4.5 Hermetic seal.-The method and procedure for performing the hermetic seal test shall be as approved by the Government.

4.4.6 Functional

4.4.6.1 Ballistics.-The test sample shall be rigidly supported in the fixture and shall be tested within five (5) minutes after its removal from the conditioning chamber. When this time limit is exceeded the sample shall be reconditioned in accordance with 4.4.6.1.1.

4.4.6.1.1 Temperature conditioning.-A standard JAU-3/A Initiator loaded with dummy IIA pellets and a dummy M42 primer and delay element containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.-Packing and marking shall be as specified by the procuring activity: Level A - Dwgs. D9280891 and D9280892, Level B - Dwgs. D9280891 and D9280893.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

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6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure time record.

6.4 Explosive Hazard Classification Information:

Explosive Hazard Classification (Quantity Distance) Class 1
Storage Compatibility - Group B, E or N
Department of Transportation (DOT) Hazard - Class C
DOT Marking - "Explosive Power Device - Class C - Handle Carefully, Keep Fire Away."

6.5 Hazard notice.-The initiator described herein and certain of its components are flammable and/or explosive, consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

6.6 If a single contractor manufactures the total item, the classification of defects does not apply. The appropriate specification shall be used.

Custodian:

Army - MU

Preparing activity:

Army - MU(FA)

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